

Notations :

- 1.Options shown in green color and with ✓ icon are correct.
- 2.Options shown in red color and with ✗ icon are incorrect.

Change Font Color :	No
Change Background Color :	No
Change Theme :	No
Help Button :	No
Show Reports :	No
Show Progress Bar :	No

Industry 4.0

Section Id :	640653139218
Section Number :	1
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	22
Number of Questions to be attempted :	22
Section Marks :	45
Display Number Panel :	Yes
Section Negative Marks :	0
Group All Questions :	No
Section Maximum Duration :	0
Section Minimum Duration :	0
Section Time In :	Minutes
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	640653389967
Question Shuffling Allowed :	No

Question Number : 1 Question Id : 6406532136119 Question Type : MCQ

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DEGREE LEVEL : INDUSTRY 4.0 (COMPUTER BASED

EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?

CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

6406536957441. ✓ Yes

6406536957442. ✗ No

Sub-Section Number :

2

Sub-Section Id :

640653389968

Question Shuffling Allowed :

No

Question Id : 6406532136120 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (2 to 4)

Question Label : Comprehension

Suppose you want to invest \$4000 now and \$2000 at the start of years 2 to 4. The interest rate offered by First Bank is 8% compounded annually, and the bonuses over the next 4 years are 1.8%, 1.7%, 2.1% and 2.5%, respectively. The annual interest rate offered by Second Bank is 0.2% lower than that of First Bank, but its bonus is 0.5% higher. The objective is to maximize the accumulated capital at the end of 4 years. If you apply Dynamic Programming to this problem, answer the given subquestions:

Sub questions

Question Number : 2 Question Id : 6406532136121 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

How is a stage represented by?

Options :

6406536957443. ✓ Year

6406536957444. ✗ Capital available for investment at the start of a year

6406536957445. ✗ Amounts invested in First Bank

6406536957446. ✗ Amounts invested in First Bank and Second Bank

Question Number : 3 Question Id : 6406532136122 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

What are the alternatives in a stage?

Options :

6406536957447. ✗ Year

6406536957448. ✗ Capital available for investment at the start of a year

6406536957449. ✖ Amounts invested in First Bank in a year

6406536957450. ✔ Amounts invested in First Bank and Second Bank in a year

Question Number : 4 Question Id : 6406532136123 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

How state variable is defined for this problem?

Options :

6406536957451. ✖ Year

6406536957452. ✔ Capital available for investment at the start of a year

6406536957453. ✖ Amounts invested in First Bank in a year

6406536957454. ✖ Amounts invested in First Bank and Second Bank in a year

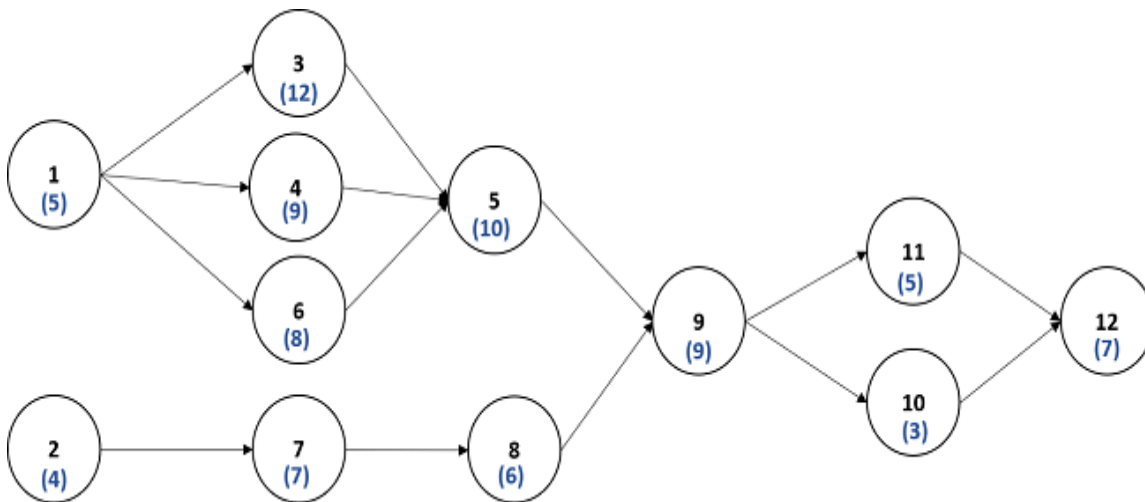
Question Id : 6406532136127 Question Type : COMPREHENSION Sub Question Shuffling

Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (5 to 7)

Question Label : Comprehension

Consider the following precedence diagram. It has 12 activities. The numbers in blue and within the braces denote the respective activity times in minutes. Answer the 3 subquestions based on the given diagram.



Sub questions

Question Number : 5 Question Id : 6406532136128 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 1

Question Label : Short Answer Question

What is the minimum cycle time (in minutes) of the system?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

12

Question Number : 6 Question Id : 6406532136129 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 1

Question Label : Short Answer Question

What is the maximum cycle time (in minutes) of the system?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

85

Question Number : 7 Question Id : 6406532136130 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

The factory works for one 8-hour shift each day. Is it possible to achieve an output of 32 units/day

Options :

6406536957459. ✓ Yes

6406536957460. ✗ No

Question Id : 6406532136131 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (8 to 9)

Question Label : Comprehension

Suppose you need to accept reservations for a flight. There are a total of 100 seats on the aircraft. You have two classes of passengers: economy and business. An economy seat generates ₹2,500 in revenue, while a business seat generates ₹7,500. Based on your experience, the demand for business travellers follows a uniform distribution between 10 and 20. Answer the given subquestions:

Sub questions

Question Number : 8 Question Id : 6406532136132 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 1.5

Question Label : Short Answer Question

What is the Booking limit?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

17

Question Number : 9 Question Id : 6406532136133 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 1.5

Question Label : Short Answer Question

What is the protection level?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

83

Sub-Section Number :

3

Sub-Section Id :

640653389969

Question Shuffling Allowed :

No

Question Id : 6406532136124 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (10 to 11)

Question Label : Comprehension

Ramu puts up a stall for Diwali Light (Diya) every year. He builds a Diya at a cost of Rs. 3 and earns Rs.5 by selling each Diya. Any Diya not sold by Diwali will be discounted to Rs. 1. From his experience, Ramu knows that the demand is Uniformly Distributed within [800, 1000] units. Then answer the given sub-questions:

Sub questions

Question Number : 10 Question Id : 6406532136125 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 1

Question Label : Short Answer Question

What is the Critical Ratio?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

0.5

Question Number : 11 Question Id : 6406532136126 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 1

Question Label : Short Answer Question

What is the optimal order quantity that minimizes Ramu's total cost?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

895 to 900

Sub-Section Number : 4

Sub-Section Id : 640653389970

Question Shuffling Allowed : No

Question Id : 6406532136134 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (12 to 14)

Question Label : Comprehension

A fashion retailer sells a limited-edition jacket for winter. The demand is normally distributed with a mean of 300 units and a standard deviation of 50 units. The cost price of each jacket is ₹1,500, and the selling price is ₹3,000. Unsold jackets can be returned for ₹1,200 per unit.

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 12 Question Id : 6406532136135 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 1

Question Label : Short Answer Question

What is the critical ratio? Round off your answer to two decimals.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

0.82 to 0.84

Question Number : 13 Question Id : 6406532136136 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 1

Question Label : Short Answer Question

What is the optimal order quantity (z-value= 0.97)

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

347 to 349

Question Number : 14 Question Id : 6406532136137 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 2

Question Label : Short Answer Question

If the retailer orders 350 jackets, what is the probability of a stockout? Enter your answer to three decimals in numbers (not in percentage). E.g.if your answer is 18.34%, enter the answer as 0.183 (Norm.s.dist for this z-value= 0.8413)

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

0.155 to 0.164

Sub-Section Number : 5

Sub-Section Id : 640653389971

Question Shuffling Allowed : Yes

Question Number : 15 Question Id : 6406532136138 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 2

Question Label : Short Answer Question

Colony Shop is stocking heavy coats for next winter. Colony pays \$50 for coat and sells it for \$110. At the end of the winter season, Colony Shop offers the coats at \$55 each. The demand for coats during the winter season is more than 20 but less than or equal to 30, all with equal probabilities. Determine the optimal order quantity that will maximize revenue for Colony Shop. Consider demand to be continuous.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Range
Text Areas : PlainText
Possible Answers :
29 to 33

Question Number : 16 **Question Id :** 6406532136139 **Question Type :** SA **Keyboard Layout :** Inscript
Correct Marks : 2

Question Label : Short Answer Question
Find the Euclidean distance between the points P(2, -1, 5) and Q(-3, 4, 2) in 3D space.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Range
Text Areas : PlainText
Possible Answers :
7.5 to 8

Sub-Section Number : 6
Sub-Section Id : 640653389972
Question Shuffling Allowed : Yes

Question Number : 17 **Question Id :** 6406532136140 **Question Type :** MSQ
Correct Marks : 3 **Max. Selectable Options :** 0

Question Label : Multiple Select Question
Select the appropriate purposes of facility layout from the list below:

Options :
6406536957468. ✔ minimize delay in material handling
6406536957469. ✔ enhance sales as appropriate in manufacturing and service facilities.
6406536957470. ✔ Provide for good maintenance
6406536957471. ✖ Funding a new project

Sub-Section Number : 7
Sub-Section Id : 640653389973
Question Shuffling Allowed : Yes

Question Number : 18 **Question Id :** 6406532136141 **Question Type :** MSQ
Correct Marks : 2 **Max. Selectable Options :** 0

Question Label : Multiple Select Question

Which of the following is/are considered in the Vehicle Routing Problem but not in TSP?

Options :

- 6406536957472. ✖ Subtour elimination
- 6406536957473. ✔ Any capacity restrictions
- 6406536957474. ✔ A specific starting point
- 6406536957475. ✖ Distance travelled restriction

Question Number : 19 Question Id : 6406532136142 Question Type : MSQ

Correct Marks : 2 Max. Selectable Options : 0

Question Label : Multiple Select Question

Select the correct statements from below:

Options :

- 6406536957476. ✖ Optimal solution in a facility location problem using Euclidean method will result in a region
- 6406536957477. ✔ Optimal solution in a facility location problem using Euclidean method rarely matches the center of gravity
- 6406536957478. ✔ Optimal solution in a facility location problem using Euclidean method will result in a point
- 6406536957479. ✖ Optimal solution in a facility location problem using Euclidean method matches the centre of gravity in not of the cases

Question Number : 20 Question Id : 6406532136143 Question Type : MSQ

Correct Marks : 2 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following example(s) can be referred to as seasonal pattern?

Options :

- 6406536957480. ✖ Economic recession
- 6406536957481. ✔ Increases Ceiling Fan sales during winters
- 6406536957482. ✔ Increased tourist footfall towards the end of the year

Question Number : 21 Question Id : 6406532136144 Question Type : MSQ

Correct Marks : 2 Max. Selectable Options : 0

Question Label : Multiple Select Question

Identify the examples for Outsourcing from the list below:

Options :

- 6406536957483. ✔ Concentrix manages customer service for Amazon .
- 6406536957484. ✖ Mastercard has set up a Data Science team in Gurgaon who will develop algorithm for Mastercard.
- 6406536957485. ✔ IIT Madras BS degree program has assigned TCS to create and manage the exam portal and its evaluation
- 6406536957486. ✖ Micron Technology, a US based semiconductor giant, setting up a Supply

Chain Analytics team, in India.

Question Number : 22 Question Id : 6406532136145 Question Type : MSQ

Correct Marks : 2 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following supply chain costs is/are affected by distribution network structure?

Options :

6406536957487. ✓ Inventory

6406536957488. ✓ Transportation

6406536957489. ✓ Facilities and handling

6406536957490. ✓ Information

6406536957491. ✗ Price

Sub-Section Number :

8

Sub-Section Id :

640653389974

Question Shuffling Allowed :

Yes

Question Number : 23 Question Id : 6406532136146 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

What is First Mile in a supply chain?

Options :

6406536957492. ✓ Movements from Vendor to Distribution centres

6406536957493. ✗ Movements from Distribution centres to stores

6406536957494. ✗ Movements from stores to customers

Question Number : 24 Question Id : 6406532136149 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Identify the Industry 4.0 version of transportation.

Options :

6406536957503. ✗ IC engine

6406536957504. ✗ Electric car

6406536957505. ✓ Hyperloop

Sub-Section Number :

9

Sub-Section Id :

640653389975

Question Shuffling Allowed :

Yes

Question Number : 25 Question Id : 6406532136147 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

Match Column A with Column B (country of origin)

Column A	Column B
1 st revolution	England
2 nd revolution	USA
3 rd revolution	USA/Japan
4 th revolution	Germany

Options :

6406536957495. ✓ A1:B1, A2:B2, A3:B3, A4:B4

6406536957496. ✗ A1:B2, A2:B1, A3:B3, A4:B4

6406536957497. ✗ A1:B3, A2:B1, A3:B4, A4:B2

6406536957498. ✗ A1:B3, A2:B1, A3: B2, A4:B4

Question Number : 26 Question Id : 6406532136148 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

Select the layout that do not require general supervision:

Options :

6406536957499. ✗ Product layout

6406536957500. ✓ Process layout

6406536957501. ✗ Group layout

6406536957502. ✗ Fixed position layout

Question Number : 27 Question Id : 6406532136150 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

What is the name of the forecasting method represented by the following set of equations? The notations have their usual meaning.

$$F_{t+1} = a_t + b_t$$

Where

$$a_t = \alpha D_t + (1 - \alpha)(a_{t-1} - b_{t-1});$$

$$b_t = \beta(a_t - a_{t-1}) + (1 - \beta)b_{t-1}$$

Options :

6406536957506. ✗ Exponential smoothing

6406536957507. ✗ Winter's model for seasonality

6406536957508. ✓ Holt's method for trend

6406536957509. ✗ None of these

Question Number : 28 Question Id : 6406532136153 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

Which of the following is not an example of digitization?

Options :

6406536957518. ✖ Creating the e-copy of a book using Google-Lens

6406536957519. ✔ Generating an automated daily sales report across the retail outlets in a state.

6406536957520. ✖ Converting a physical 2D sheet to CAD drawing

6406536957521. ✖ Converting physical invoice to e-copy

Sub-Section Number : 10

Sub-Section Id : 640653389976

Question Shuffling Allowed : Yes

Question Number : 29 Question Id : 6406532136151 Question Type : MCQ

Correct Marks : 1.5

Question Label : Multiple Choice Question

Which of the following functions does Logistics Management not cover?

Options :

6406536957510. ✖ Supply Management

6406536957511. ✖ Materials Management

6406536957512. ✔ Customer Management

6406536957513. ✖ Distribution Management

Question Number : 30 Question Id : 6406532136152 Question Type : MCQ

Correct Marks : 1.5

Question Label : Multiple Choice Question

Which of the following is not a form of internet?

Options :

6406536957514. ✖ Industrial

6406536957515. ✖ Commercial

6406536957516. ✔ Philanthropic

6406536957517. ✖ Consumer